React Introduction

What is React?

React History

- Created by a software engineer at Facebook
- Later used on Instagram
- Open-sourced in 2013

React is a JavaScript library

- React uses JavaScript
- Code is executed in the user’s own browser
- Fast, modern web apps

React History

- Challenged established best practice
  - MVC (Model View Controller)
  - Templating
- React only concerned with ‘V’ (the View)
- Focused on the visual components of a UI

Single Page Applications (SPAs)

- React allows us to create a single-page application
- Page is rendered once and the remainder of the user’s interactions are executed in the browser
- React components can also be used in multi-page applications

React code

- Looks like a mixture of JavaScript and HTML

```
function App() {
  const [count, setCount] = useState(0)

  return (
    <>
      <div>
        <a href="https://vite.dev" target="_blank">
          <img src={viteLogo} className="logo" alt="Vite logo" />
        </a>
        <a href="https://react.dev" target="_blank">
          <img src={reactLogo} className="logo react" alt="React logo" />
        </a>
      </div>
      <h1>Vite + React</h1>
```

- Is actually “JSX” code (JavaScript XML)

JSX

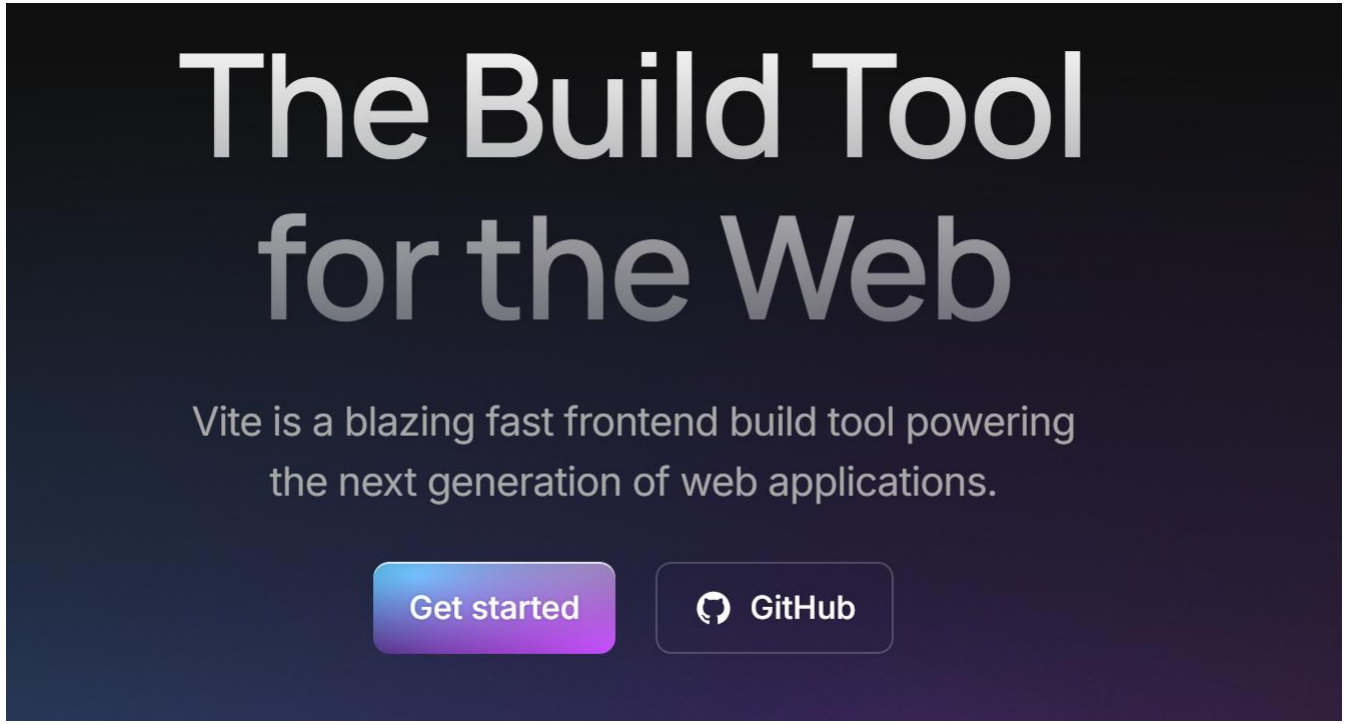
- JSX stands for JavaScript XML
- JSX allows us to write HTML-like code inside React components
- Needs to be converted to standard JavaScript in order to be rendered by a web browser

Development environment

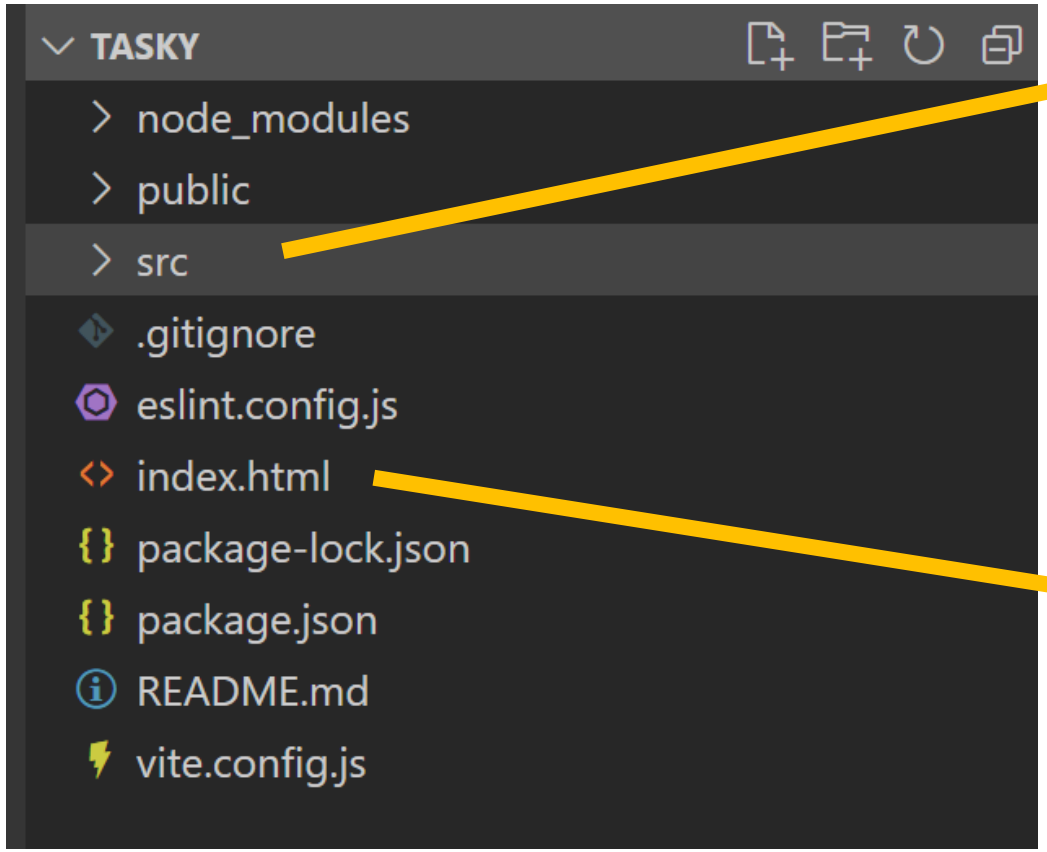
- Node.js
- npm (node package manager)
- Git / GitHub
- Visual Studio Code
- Chrome
- Vite

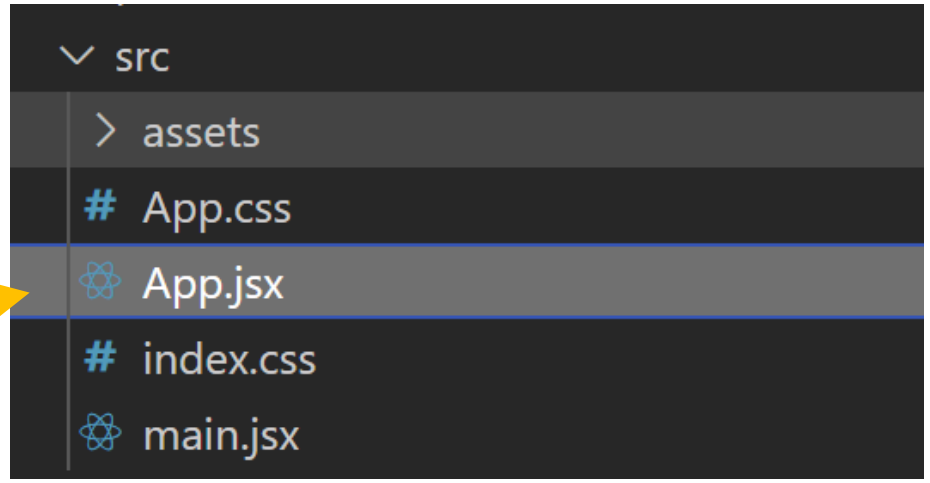
Vite

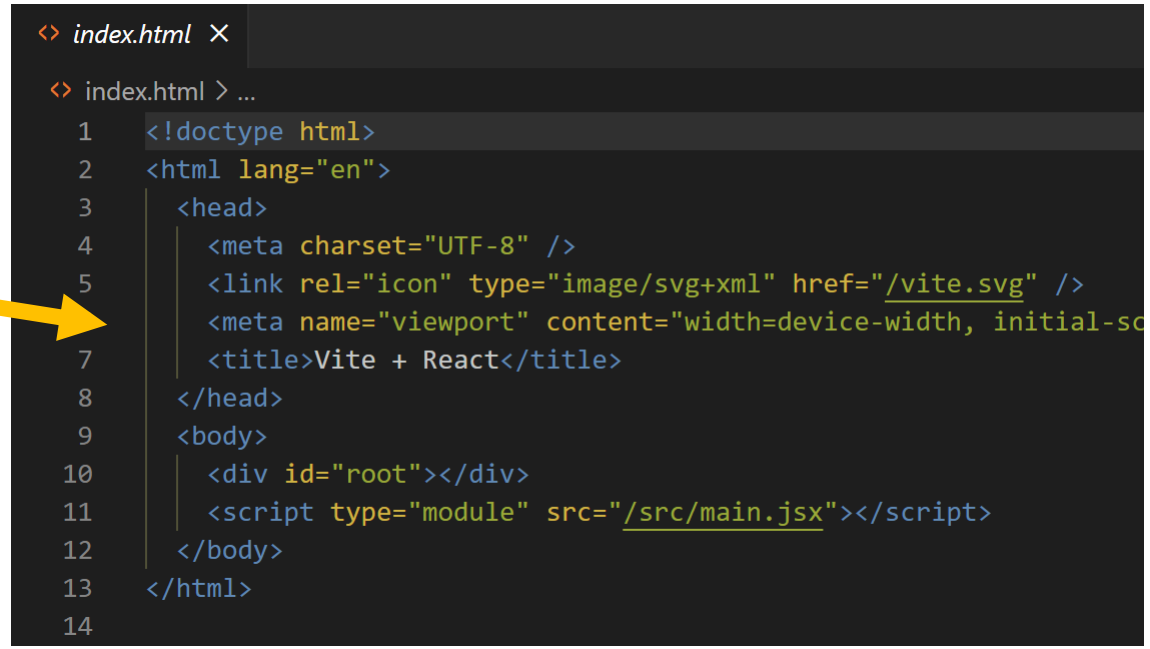
- Vite is a build tool that creates starter apps for front-end frameworks such as React, Vue, Svelte, and others



Test app folder structure







Vite

- When we create a React app with Vite, it installs and configures all the packages needed to:
  - Use the React library
  - Use the react-dom library (for interactions with the browser DOM)
  - Transpile JSX code so that it can be interpreted by any browser
  - Restart the development server each time you make changes to the code
  - And lots more!